

Social Media Text Normalization

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1. Introduction

Social media platforms generate vast amounts of text which is generated by the users, often written in an informal and a bit casual manner. Due to the presence of spelling variations, code switching, abbreviations and emojis, processing this data is a challenge for Natural Language Processing tasks. Our project will try to develop a **Text Normalization Pipeline** that normalizes social media content in Hindi (Hinglish) language. By addressing the inconsistencies in a social media user-generated text, this pipeline will enhance the pre-existing applications of NLP such as sentiment analysis, hate speech detection, and text classification.

2. Problem Statement

The main challenge in processing social media text is its lack of standardization, and a lot of inconsistencies in it which arises due to:

- **Spelling variations and errors:** Misspellings, phonetic substitutions, and typos.
- **Transliteration issues:** Roman script usage for Indian languages.
- **Code-mixing and code-switching:** Frequent switching between English and native languages.
- **Abbreviations and shorthand:** Use of contractions and internet slang.
- **Emojis and non-standard punctuation:** Affecting text semantics.

These irregularities make it difficult for NLP models to effectively interpret social media content. Our objective is to build a normalization framework that mitigates these issues, enhances data quality and build a social media text normalization NLP pipeline.

3. Scope of the Project

This project will focus on:

- Normalizing text for **Hindi** (Hinglish).
- Collecting a **corpus of real-world social media posts** for language-specific analysis.
- Developing a **comprehensive normalization pipeline** that addresses:
 - Spelling correction
 - Transliteration normalization
 - Handling code-switching and code-mixing
 - Expanding abbreviations
 - Processing emojis and special characters
- Implementing **rule-based and statistical approaches** for normalization.
- Creating a **gold-standard dataset** with annotated normalized text.
- Evaluating the impact of normalization on NLP tasks like **sentiment analysis and hate speech detection**.

- **Exploring Hate Speech Detection:** If time permits, we will also work on detecting hate speech in social media text, analyzing its relation to emojis and other informal expressions.
- The scope may be updated to include additional aspects or remove certain elements based on feasibility and project progress.

This project will significantly improve text preprocessing techniques for low-resource Indian languages in NLP applications.

4. Datasets

Finding the right dataset is crucial for our project. While we are still exploring the best-fit datasets, we have found the following relevant resources for now:

4.1. PHINC: Parallel Hinglish Social Media Code-Mixed Corpus

- **Description:** Contains **13,738** Hindi-English code-mixed sentences from social media, each paired with human-translated English sentences.
- **Usefulness:** Ideal for studying **code-mixing patterns** and training **text normalization models**.
- **Access:** Available on ARXIV.

4.2. L3Cube-HingCorpus

- **Description:** A **large-scale corpus** containing **52.93 million sentences** and **1.04 billion tokens** of Hindi-English code-mixed text collected from Twitter.
- **Usefulness:** Suitable for **pre-training language models** and developing **text normalization pipelines**.
- **Access:** Available on ARXIV.

4.3. Aggression-Annotated Corpus of Hindi-English Code-Mixed Data

- **Description:** Comprises approximately **18,000 tweets** and **21,000 Facebook comments** annotated for aggression in Hindi-English code-mixed language.
- **Usefulness:** Useful for analyzing **informal language** and **offensive content** in social media.
- **Access:** Available on ARXIV.

4.4. Hindi-English Code-Mixed Corpus for Hate Speech Detection

- **Description:** A dataset developed for **hate speech detection** in Hindi-English code-mixed social media text, consisting of tweets annotated for hate speech.
- **Usefulness:** Beneficial for studying **normalization's impact** on **hate speech detection**.
- **Access:** Available on SpringerLink.

4.5. BHAAV: Hindi Text Corpus for Emotion Analysis

- **Description:** Contains **20,304 sentences** from **230 Hindi short stories** across 18 genres, annotated for emotions like *anger*, *joy*, *suspense*, *sadness*, and *neutrality*.
- **Usefulness:** Valuable for understanding **emotional expressions** in Hindi text.
- **Access:** Available on GitHub.

4.6. Additional Hate Speech Dataset

- **Description:** A dataset specifically designed for **hate speech detection** in Hindi-English code-mixed text.
- **Usefulness:** Helps in identifying hate speech patterns and evaluating the effectiveness of normalization.
- **Access:** Available at ACL Anthology.

Note: These datasets are not final. We are still in the process of identifying the most suitable datasets for our project. Depending on feasibility, availability, and project requirements, we may modify this selection by adding new datasets or replacing the current ones.

5. Evaluation Metrics

To effectively evaluate the performance of the proposed text normalization pipeline, we will employ multiple quantitative and qualitative metrics. The evaluation process will focus on measuring the accuracy, fluency, and generalization capabilities of the pipeline across different Indian languages commonly used on social media. The following metrics will be utilized:

5.1. Edit Distance

Edit Distance, also known as Levenshtein Distance, measures the minimum number of single-character edits (insertions, deletions, substitutions) required to transform the original text into the normalized text. A lower edit distance indicates higher similarity and better normalization quality. This metric is particularly useful for evaluating spelling correction, transliteration consistency, and abbreviation expansion.

- **Formula:**

$$d(i, j) = \min \begin{cases} d(i-1, j) + 1, \\ d(i, j-1) + 1, \\ d(i-1, j-1) + 1 \quad \text{if } x_i \neq y_j \end{cases}$$

- **Application:** Useful in assessing spelling correction and transliteration consistency.

5.2. BLEU Score (Bilingual Evaluation Understudy)

BLEU Score is an n-gram based metric traditionally used for evaluating machine translation, but it can be effectively applied to normalization tasks by comparing the generated output with a human-annotated reference. Higher BLEU scores indicate greater similarity to the reference standard.

- **Formula:**

$$\text{BLEU} = BP \times \exp \left(\sum_{n=1}^N w_n \log p_n \right)$$

- BP = Brevity Penalty
- p_n = Precision for n-grams
- w_n = Weights assigned to different n-gram lengths

- **Application:** Measures fluency and accuracy by comparing normalized text to a gold standard corpus.

5.3. Impact on Downstream NLP Tasks

Normalization quality is also measured by evaluating the performance improvement in NLP tasks using the normalized text compared to raw input.

- **Sentiment Analysis:** Compare classification accuracy, F1-score, precision, and recall on normalized vs. unnormalized text.

- **Named Entity Recognition (NER):** Measure improvements in detecting entities within noisy text after normalization.
- **Text Classification:** Assess changes in model performance when training on normalized text.
- **Application:** Provides insights into the practical benefits of normalization by quantifying its effect on real-world NLP tasks.

5.4. Human Evaluation (Optional)

For a qualitative assessment, human annotators can be employed to rate the normalized outputs based on:

- **Fluency:** The readability and grammatical correctness of the normalized text.
- **Adequacy:** The extent to which the normalized text retains the intended meaning of the original text.
- **Application:** Useful for evaluating nuanced language variations, especially for code-switching and mixed-script inputs.

6. Literature Review

Existing research in text normalization has demonstrated its crucial role in enhancing NLP performance by reducing data sparsity, improving model generalization, and ensuring consistency across different language inputs. Previous studies have employed various approaches, primarily focusing on rule-based systems, statistical models, deep learning techniques, and hybrid methods.

6.1. Rule-Based Approaches

Rule-based normalization systems rely on predefined rules, patterns, and lexicons to standardize text. These methods are particularly effective for handling common spelling errors, abbreviation expansion, and transliteration.

- **Example:** Systems that map commonly used slang or shorthand (e.g., “u” → “you”) to their standard forms.
- **Strengths:** High accuracy for predefined rules and language-specific corrections.
- **Limitations:** Inflexible to evolving language trends and unable to generalize beyond known patterns.

6.2. Statistical Models

Statistical approaches leverage probabilistic models to perform text normalization by learning patterns from annotated corpora.

- **Example:** Noisy Channel Model, where the normalization task is framed as finding the most likely original text given a noisy input.
- **Strengths:** Can generalize beyond predefined rules and accommodate spelling variations.
- **Limitations:** Requires large, annotated datasets for training; may fail in low-resource scenarios.

6.3. Neural Network-Based Approaches (Seq2Seq Models)

Sequence-to-sequence (Seq2Seq) models with attention mechanisms have emerged as a popular approach for text normalization. These models are capable of handling complex input-output mappings and generating normalized text directly from noisy input.

- **Example:** Transformer-based architectures have been applied to normalize social media text with promising results.

- **Strengths:** Capable of learning deep representations of text, allowing for better generalization across various domains.
- **Limitations:** Requires substantial annotated data and computational resources; susceptible to overfitting in low-resource scenarios.

6.4. Hybrid Approaches

Hybrid models combine rule-based and statistical or deep learning techniques to leverage the strengths of both.

- **Example:** Systems that use rule-based preprocessing to handle straightforward cases (e.g., abbreviation expansion) before applying Seq2Seq models for more complex tasks.
- **Strengths:** Improved performance by utilizing the advantages of each method.
- **Limitations:** Integration complexity and potential redundancy between components.

6.5. Previous Work in Indian Languages

The normalization of social media text in Indian languages has received limited attention compared to English, though several efforts have been made:

- Studies addressing **Hindi-English code-mixing** using hybrid techniques combining rule-based transliteration with neural networks.
- Research focusing on **Tamil and Bengali** for standardizing Roman script inputs.
- Evaluation frameworks designed specifically for **multi-script inputs**, which are common in social media posts.

7. Expected Deliverables

- Annotated dataset of social media text with normalized versions.
- Lexicon of common variations and their standard forms.
- Normalization pipeline implementation (both rule-based and statistical approaches).
- Evaluation report comparing different approaches.
- Analysis of language-specific challenges in normalization.

8. Conclusion

Text normalization is a crucial preprocessing step for social media text analysis. This project aims to develop a comprehensive normalization pipeline for Indian languages, contributing to improved NLP performance in downstream tasks like sentiment analysis and classification.